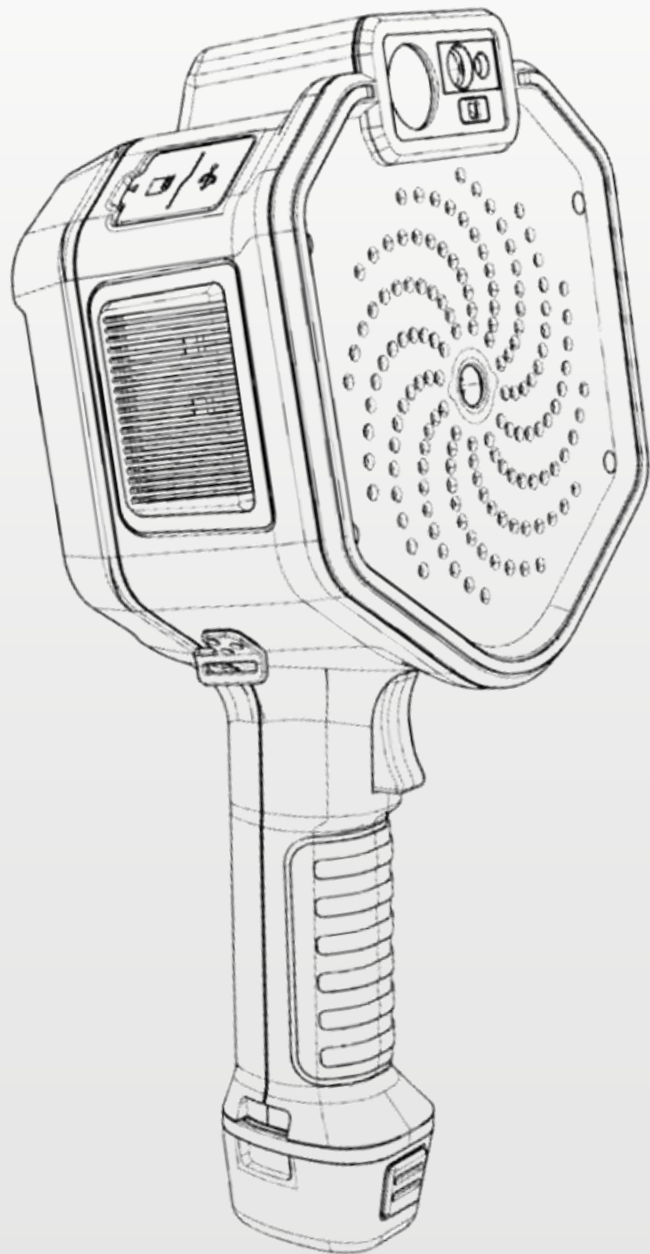


New Generations

Acoustic Cameras



What Makes Us Stand Out

What is Really Important in an Acoustic Camera?

Is it the number of microphones?

The detection bandwidth?

.....

In our latest-generation acoustic camera,
We've enhanced all the specs--
But More importantly, we focused on your real-world experience.

NEW

Features

Hardware

- 144 MEMS Microphones
- New chip for 2 ~ 130kHz
- Wi-Fi, Bluetooth, E-compass
- LED & Laser Rangefinder

Software

- AI Analysis for Partial discharge
- AI analysis for Gas leak rate
- Mechanical noise analysis
- On-device Analysis & Reported
- Spectrum display & RTSP



● Design

- Tripod Mount
- Wrist Strap
- Less than 1100g
- Removable Battery

● Improve details

- PRPD AI algorithm
- File Management
- Voice Annotation
- Video Analysis

● Battery & Charging

- 4 hours in operation mode
- 8 hours in standby mode
- 2 hours full charge

HA3

Hertzinno more professional

Acoustic

Microphones

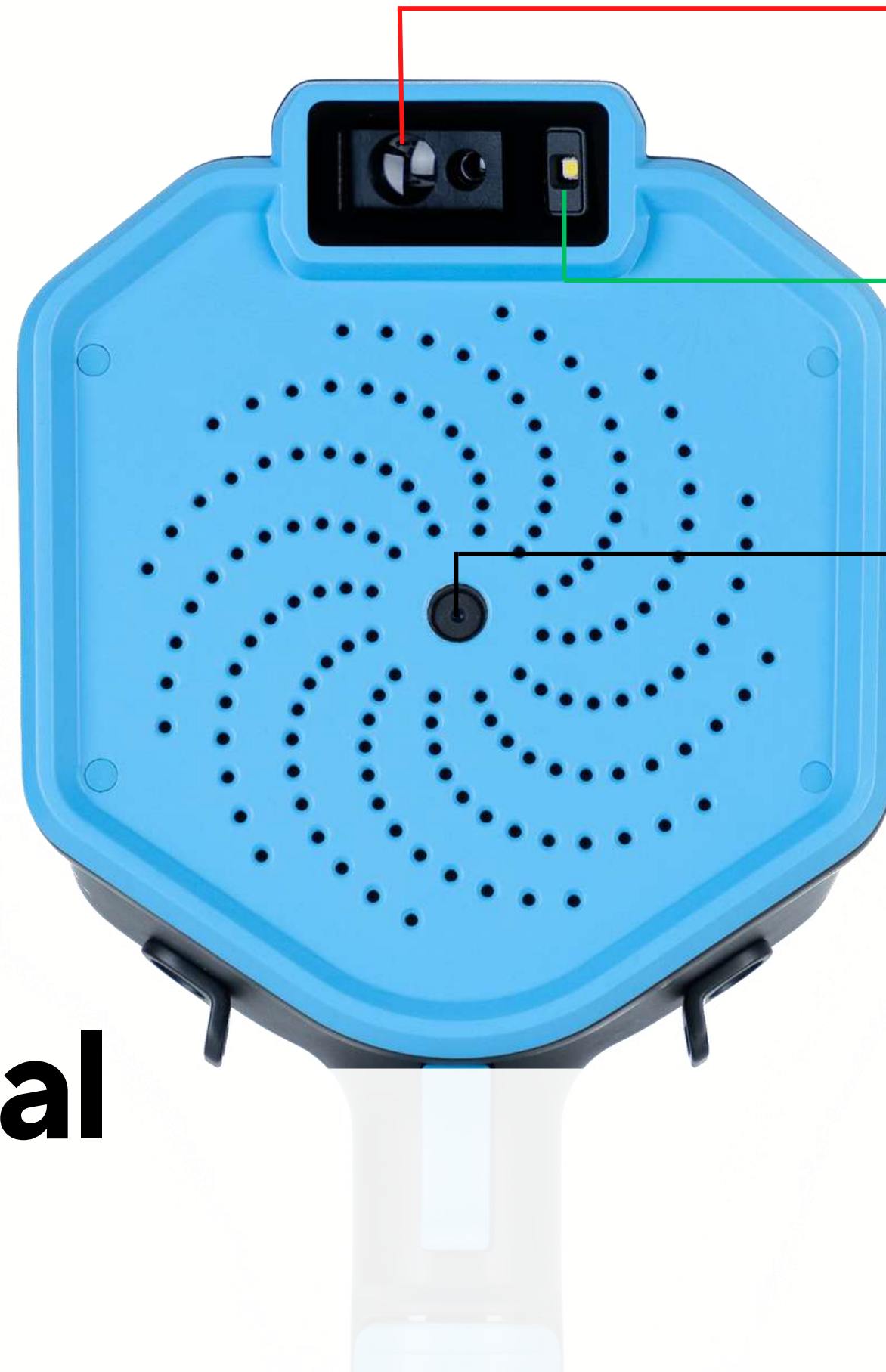
- 144 MEMS Microphone
- 2kHz ~ 130kHz

Smarter

More Powerful

- Partial Discharge Mode
- Gas Leak Detection Mode
- Mechanical Noise Mode
- Adaptive Frequency Tuning

Professional



Laser Rangefinder

Class II Laser
30 meters measure distance

LED & Storage

High-brightness LED Light 1W
Removable MicroSD Card

Cameras & Others

13MP Optical Camera
8x Digital Zoom

- Less than 1100g
- 4h Operation time
- 8 hours Standby time
- IP54

HA3T

Hertzinno more professional

Thermal

IR & ACOUSTIC

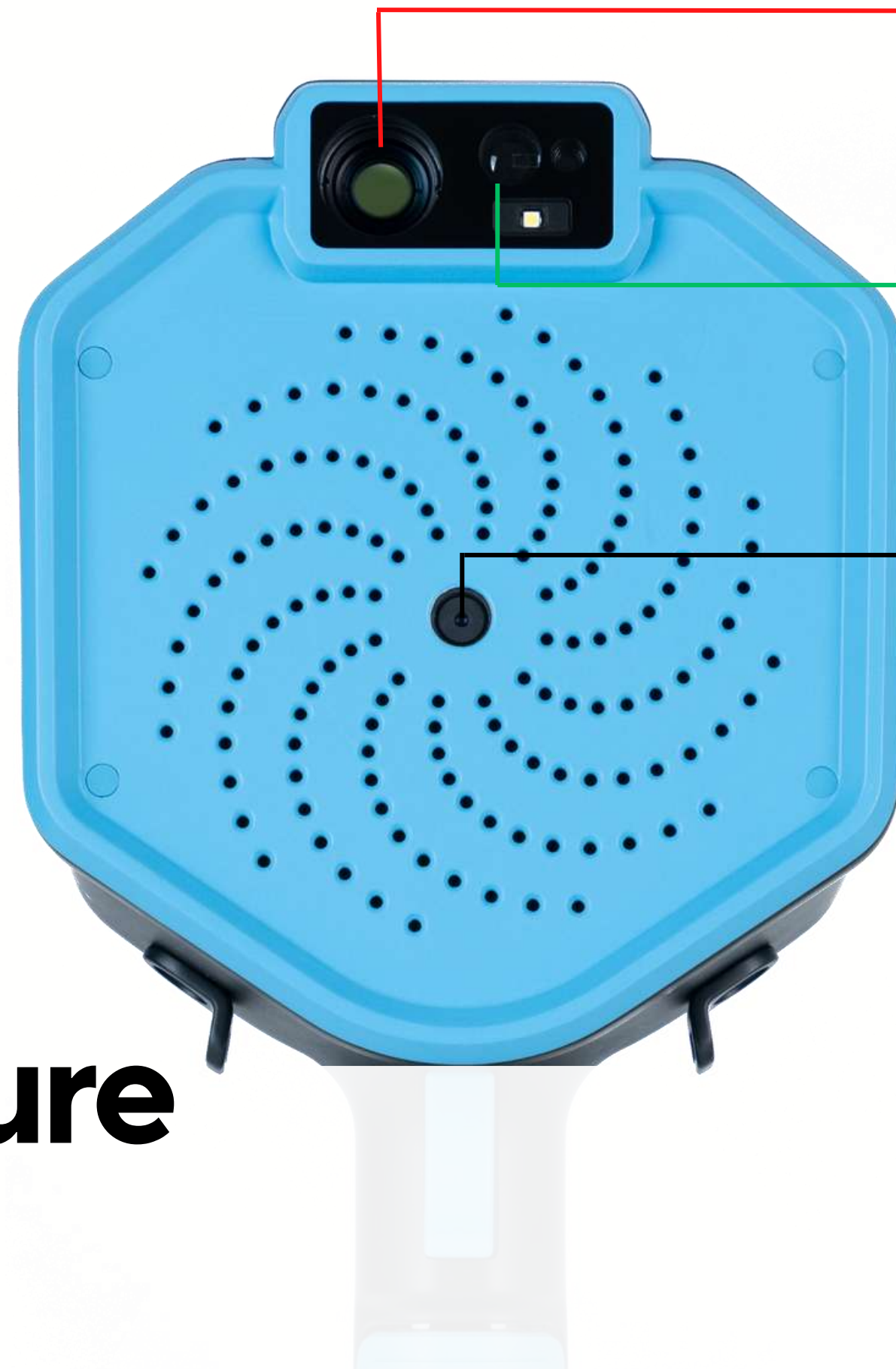
- 640*512 IR resolution
- $\leq 40\text{mK} (@25^{\circ}\text{C}, F\#=1.0)$

High Sensitivity IR

Wide Range Measure

- Response band $8\sim 14\mu\text{m}$
- 0.8 mrad
- $-20^{\circ}\text{C}\sim 150^{\circ}\text{C}$ & $100^{\circ}\text{C}\sim 550^{\circ}\text{C}$
- $\pm 2^{\circ}\text{C}$ or $\pm 2\%$ of reading
(whichever is greater)

Temperature



Thermal Graph Sensor

VOx Sensor
12 μm size & 15mm inch

Rangefinder Laser & LED

Up to 30m Laser Rangefinder
High-brightness LED Light

Cameras & Others

13MP Optical Camera
144 MEMS Microphones

- Less than 1100g
- 4h Operation time
- 8 hours Standby time
- IP54

HA3LX

Hertzinno more professional

Methane

ATEX ACOUSTIC

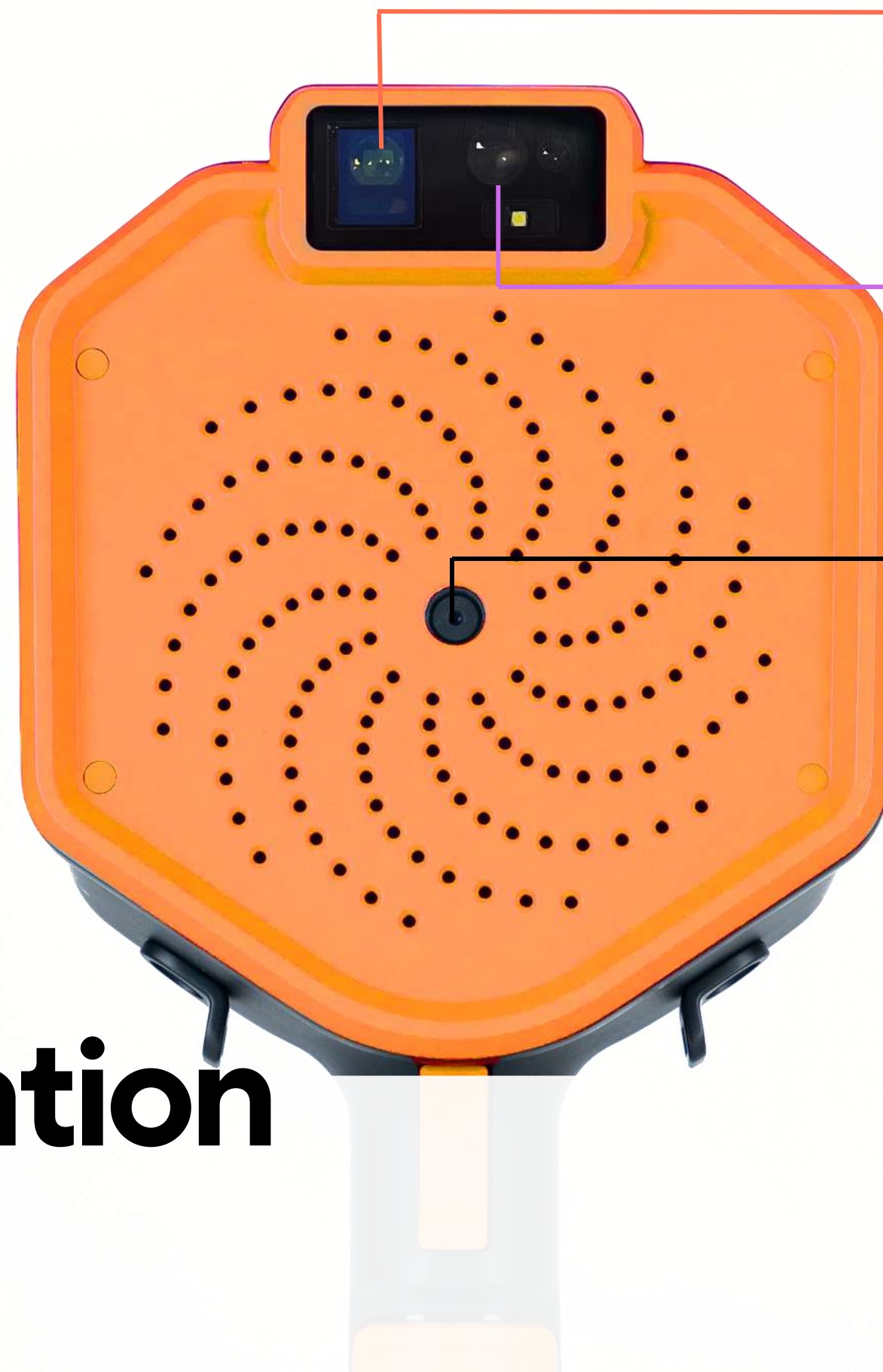
- II 2 G Ex ib IIC T4 Gb
- II 2 D Ex ib IIIC T130°C Db

Higher Sensitivity Wider Range

- -15dB SPL @20kHz
- 2 ~ 130 kHz
- <1cm@1m @40kHz
- 0 ~ 50000 ppm*m

Concentration

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● Methane Laser Sensor

5ppm*m Concentration
Leak Rate and Cost Calculate

● Rangefinder Laser & LED

Up to 30m Laser Rangefinder
High-brightness LED Light

● Cameras & Others

13MP Optical Camera
144 MEMS Microphones

- Less than 1100g
- 4h Operation time
- 8 hours Standby time
- IP54 & ATEX

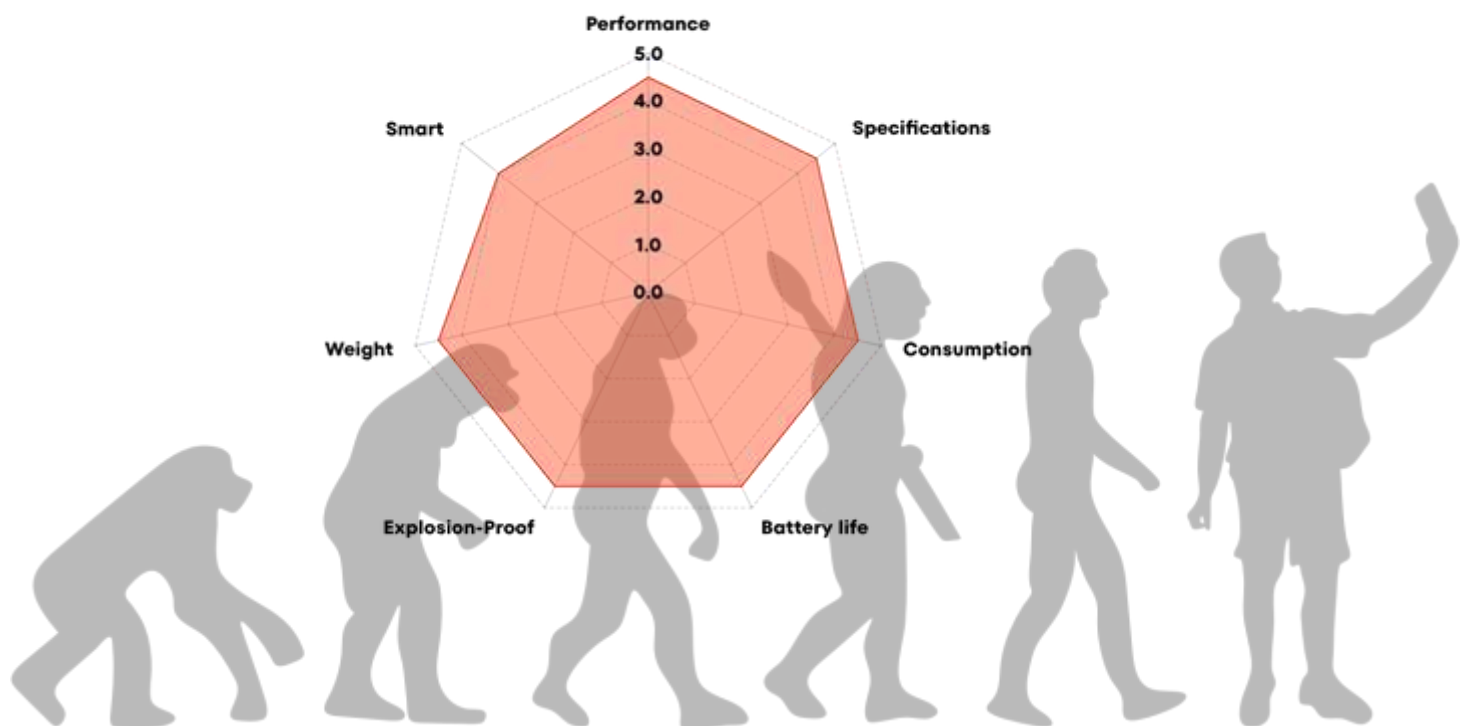
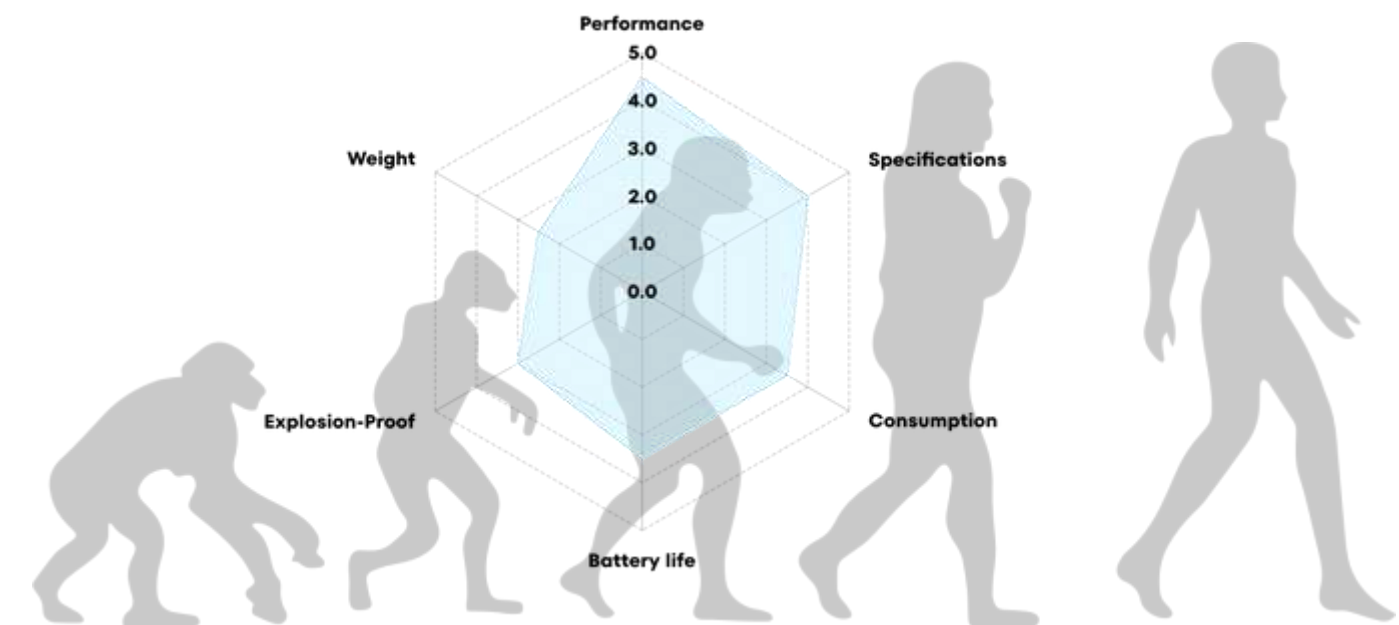
Natural Gas Leak

What`s New, What`s Better

Will the new acoustic camera replace previous models?
what is different, besides the number of sensors?
How do I choose the right acoustic camera?

.....

Designing a product always involves trade-offs — between power and portability, performance and efficiency, sensitivity and durability, and more.



Model	HZ-HA-170P/270P	HZ-HA-171P/271P	HA3/3T/3LX
Microphone	138 MEMS	128 MEMS	144 MEMS
Frequency Range	2kHz ~ 100kHz	2kHz ~ 100kHz	2kHz ~ 130kHz
Measure Distance	0.3 ~ 210m	0.3 ~ 200m	0.3 ~ 210m
Refresh Rate	30fps	30fps	30fps
Work Mode	PD, Leak, Mechanical	PD, Leak, Mechanical	PD, Leak, Mechanical
Thermal Imaging	270P	271P	HA3T
Methane Sensor	NA	NA	HA3LX
Optical Camera	13MP	13MP	13MP
Screen	5inch @ 720P	5inch @ 720P	5inch @ 1080P
LED Fill Light	NA	NA	1W
Wi-Fi & Bluetooth	Wi-Fi	Wi-Fi	Wi-Fi & Bluetooth
Other Sensor	NA	NA	E-compass, Atmospheric
Straps	Wrist Strap	NA	Neck & Wrist
Memory	64GB	64GB	128GB MicroSD Card
Battery	Internal with 4 hours	Replaceable 7 hours	Replaceable 4 hours
Laser Rangefinder	NA	NA	30m
Weight	980g	1100g	1100g
Sensitivity	10kHz: 1dB SPL 20kHz: -10dB SPL 30kHz: -6dB SPL 40kHz: 3dB SPL 50kHz: 6dB SPL 60kHz: 20dB SPL	10kHz: 1dB SPL 20kHz: -10dB SPL 30kHz: -6dB SPL 40kHz: 3dB SPL 50kHz: 6dB SPL 60kHz: 20dB SPL	10kHz: 3dB SPL 20kHz: -15dB SPL 40kHz: -5dB SPL 60kHz: 10dB SPL 80kHz: 35dB SPL 100kHz: 45dB SPL

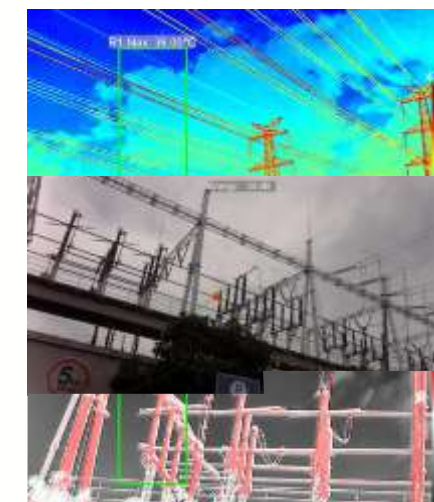
Every model represents the best choice for its specific application scenario

THE CHOICE IS SIMPLE

Scenarios

Each acoustic camera model is optimized for specific environments—from performance and functionality to weight and certification.

Choosing the right model ensures optimal performance and maximum value.



Competitive Trade Shows

HA3 series

Bluetooth, LED, 130kHz, 144

Explosive Environments

HA3LX & HA170P & HA270P

ATEX

Power Industry

HA3T & HA270P & HA271P

Acoustic & Thermal

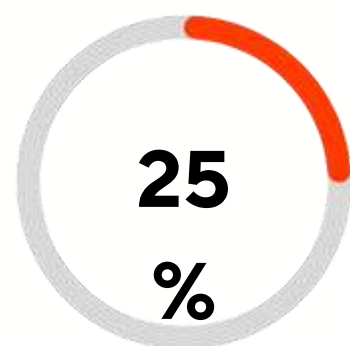
Mechanical Diagnostics

HA171P & HA271P

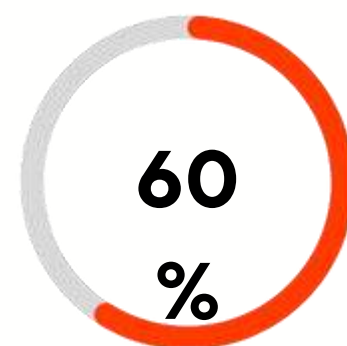
7 hours operation

Replaced battery

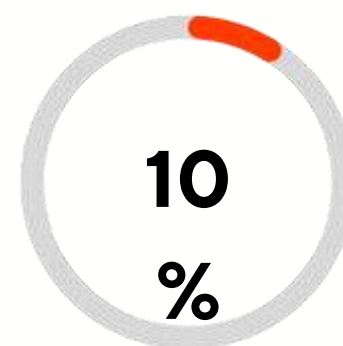
Budget



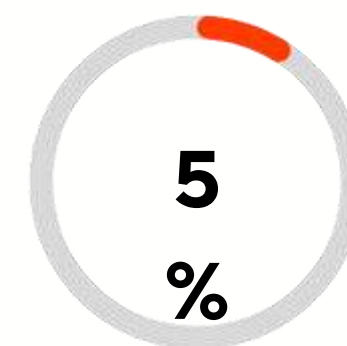
Functions



Parameters



Performance



Every model represents the best choice for its specific application
scenario

ACOUSTIC CAMERA APPLICATIONS

Power Industry

Recommended Models: HA3T & HA271P

Reasons for Recommendation:

Both models belong to our dual-sensor series, integrating **acoustic imaging** and **infrared thermography** in one device.

They are capable of detecting both capacitive and resistive faults.

Enhanced by AI recognition algorithms and multi-dimensional noise reduction of PRPD

Product Advantages:

HA3T (PD + 640*512 IR +4 hours)

Built-in Bluetooth data transmission and Bluetooth listening capabilities.

Features high-precision laser ranging.

Compact form factor with modern, ergonomic design.

HA271P (PD + 640*512 IR + 7hours)

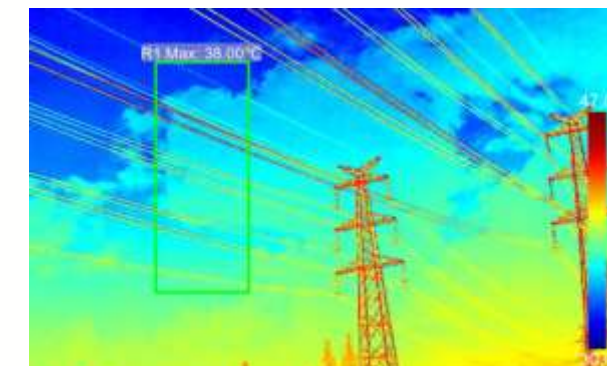
Extended battery life, with a single battery supporting up to 7 hours of operation.

Market Requirements in the Power Sector:

- Capability to detect partial discharge (PD).
- Ability to identify and analyze different PD types.
- Most use cases are outdoor environments, with a few indoor scenarios.
- Infrared imaging is commonly required alongside acoustic detection.



HA3T



HA-271P



Every model represents the best choice for its specific application
scenario

Market Requirements in the Gas Sector:

- Capable of detecting high-pressure gas leaks (including methane, hydrogen, nitrogen, etc.).
- Suitable for complex and noisy industrial environments, such as refineries and chemical plants.
- Must support safe operations in explosive zones (ATEX certified).
- Ability to visualize small leaks from long distances in outdoor scenarios.
- Often used for routine inspection and emergency maintenance.

ACOUSTIC CAMERA APPLICATIONS

Chemical Gas

Recommended Models: HA3LX & HA270P

Reasons for Recommendation:

Advanced dual-function acoustic cameras combining ultrasonic leak detection with methane laser concentration sensing (IR Sensor with 270P), delivering reliable performance in gas-leak critical environments.

Product Advantages:

HA3LX (Leak Mode + Methane Laser Sensor + ATEX)

First-in-industry ATEX-certified acoustic camera that integrates laser methane sensing, supporting real-time acoustic visualization and methane quantification. Also features Wi-Fi/Bluetooth transmission and compact design.

HA270P (Leak Mode + 640*512 IR + ATEX)

infrared imaging capability, and highly sensitive leak localization. Rugged design for heavy-duty use in harsh environments.

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HA3LX



HA-270P



Every model represents the best choice for its specific application
scenario

ACOUSTIC CAMERA APPLICATIONS

Mechanical

Market Requirements in the Mechanical via Acoustic:

- Rapid Fault Localization - pinpoint abnormal sound sources fast
- Real-Time Monitoring & Early Warning - detect anomalies before failures
- Cut Downtime & costs - prevent major breakdowns
- Identify issue type - bearing wear, gear misalignment, poor lubrication
- Trend Analysis & Condition Assessment

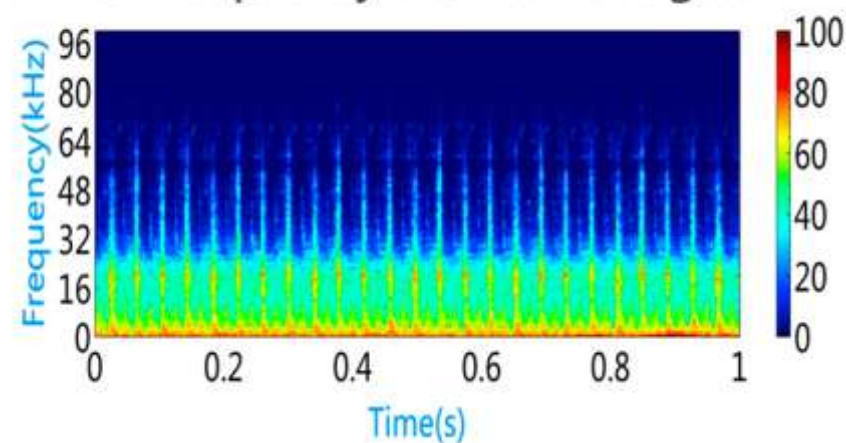
Recommended Models: All currently available HERTZINNO acoustic cameras

Reasons for Recommendation:

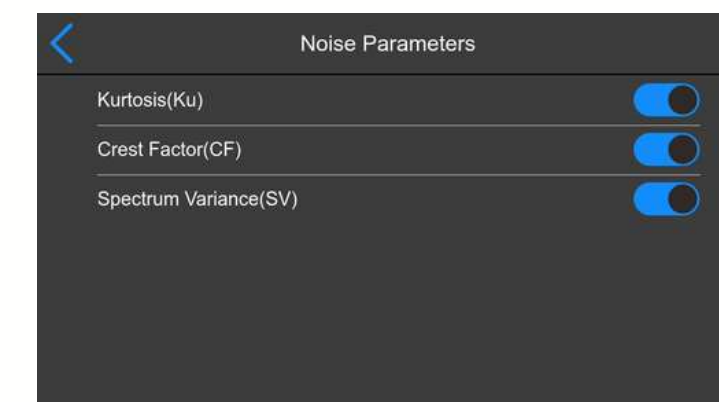
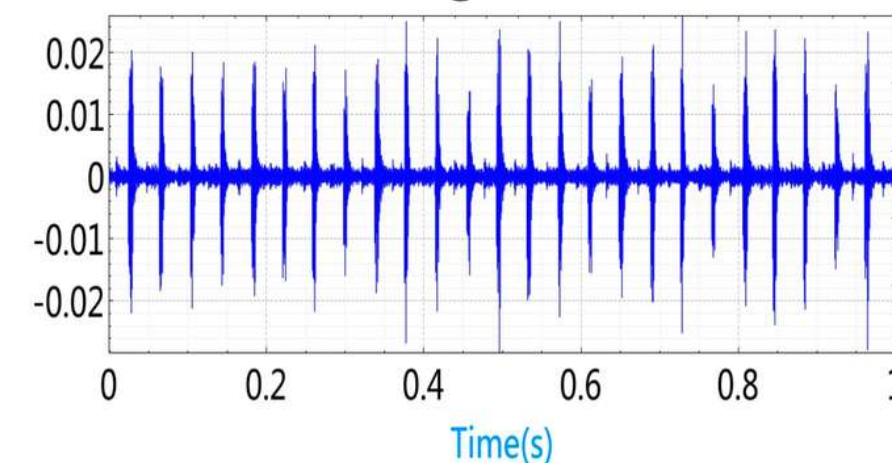
- All models are equipped with a mechanical noise detection mode
- Capable of quickly locating anomalies based on acoustic signatures



Time-Frequency Domain Diagram



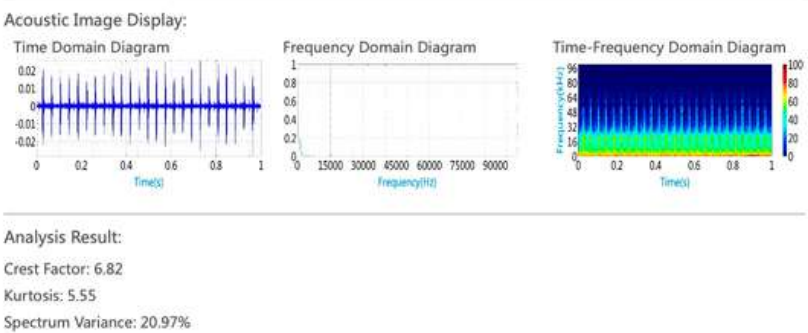
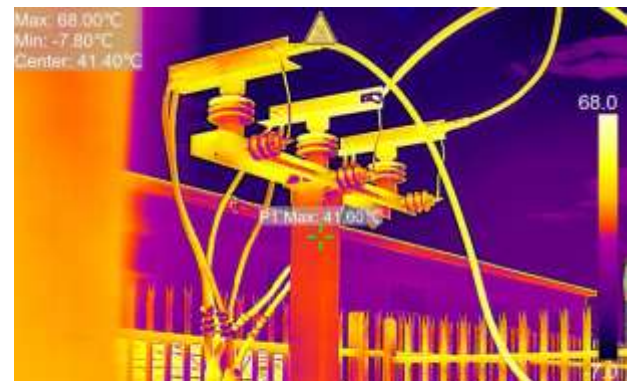
Time Domain Diagram



Designing a product always involves trade-offs — between power and portability, performance and efficiency, sensitivity and durability, and more.



Competition



Model	HA3/3T/3LX	Flir Si2Pro	Fluke ii915
Appearanc			
Microphone	144 MEMS	124 MEMS	64 MEMS
Frequency range	2kHz ~ 130kHz	2kHz ~ 130kHz	2kHz ~ 100kHz
Refresh Rate	30fps	30fps	30fps
Work Mode	PD, Leak, Mechanical	PD, Leak, Mechanical	PD, Leak, Mechanical
Thermal Imaging	HA3T	NA	NA
Methane Sensor	HA3LX	NA	NA
Optical Camera	13MP	12MP	5MP
Screen	5inch @ 1920*1080P	5inch @ 1280*720P	7inch @ 1280*800P
LED Fill Light	1W	YES	1W
Wi-Fi & Bluetooth	Wi-Fi & Bluetooth	Wi-Fi	Wi-Fi
Other Sensor	E-compass, Atmospheric	NA	NA
Straps	Neck & Wrist	Neck & Wrist	Neck & Wrist
Memory	128GB MicroSD Card	128GB SD card	32 GB
Battery(Use/Charge)	4 hours/ 2hour	2.5 hours /2.5hours	6 hours/4 hours
Laser Rangefinder	30m	NA	NA
Weight	1100g	1450g	2000g
Sensitivity	10kHz: 3dB SPL 20kHz: -15dB SPL 40kHz: -5dB SPL 60kHz: 10dB SPL 80kHz:35dB SPL 100kHz: 45dB SPL	20kHz: -7dB SPL 35kHz: 4dB SPL 50kHz: 10dB SPL 80kHz: 36dB SPL 100kHz: 51dB SPL	12.1 to 114.6 dB SPL (± 1 dB SPL 2 kHz) 4.4 to 101.2 dB SPL (± 2 dB SPL 19 kHz) 12.8 to 119.2 dB SPL (± 1 dB SPL 35 kHz) 19.8 to 116.1 dB SPL (± 3 dB SPL 52 kHz) 41.4 to 129.0 dB SPL (± 1 dB SPL 80 kHz) 54.4 to 135.5 dB SPL (± 1 dB SPL 100 kHz)



ONE MORE

Firmware

In acoustic camera comparisons, the number of microphones does not necessarily represent the overall performance of the device. Performance should be evaluated through a comprehensive assessment.

With this release, HERTZINNO has carried out a complete upgrade of the product firmware.



SMARTPHONE PORTRAIT

Firmware

Our latest software release incorporates a wide range of advanced features, including spectrum visualization, intelligent analysis of on-device measurement results, real-time screen sharing, Bluetooth headset monitoring, and ultra-high-frequency spectrum sampling.

In addition, we have further enhanced our PRPD (Phase-Resolved Partial Discharge) pattern analysis capabilities.



AI PRPD

On-device Analysis



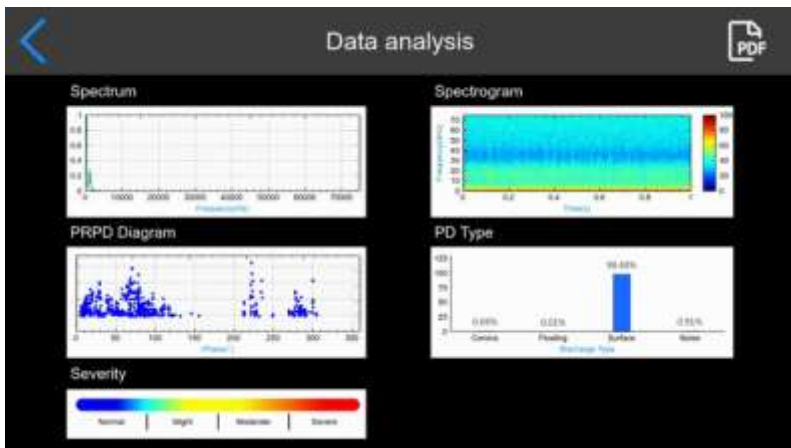
2 Mode Sound

RTSP

Firmware



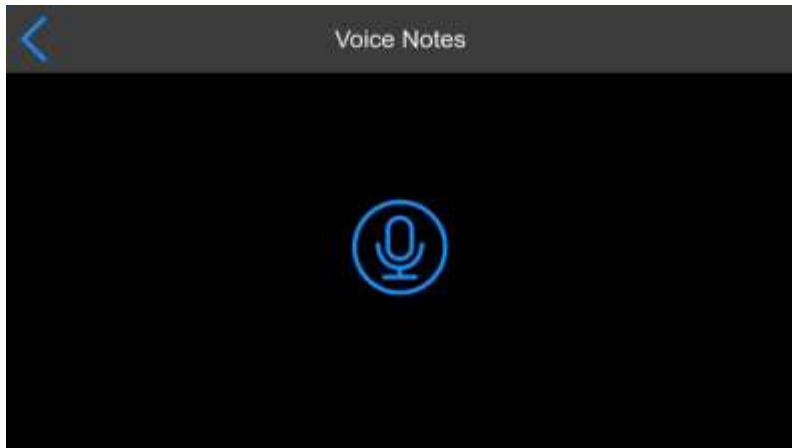
Take Photo



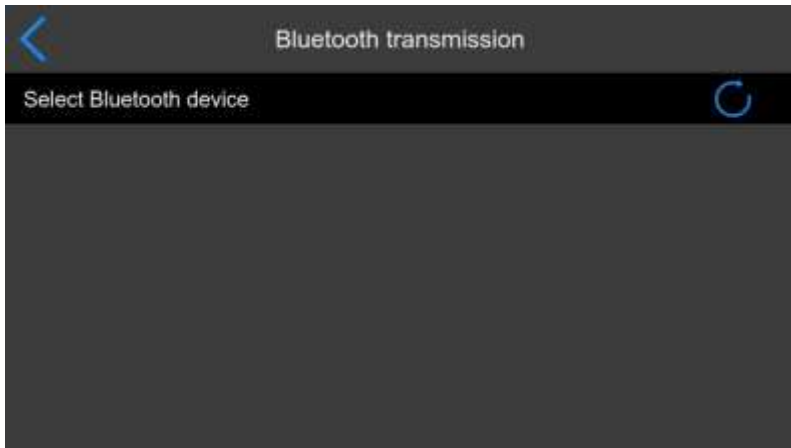
Data Analysis



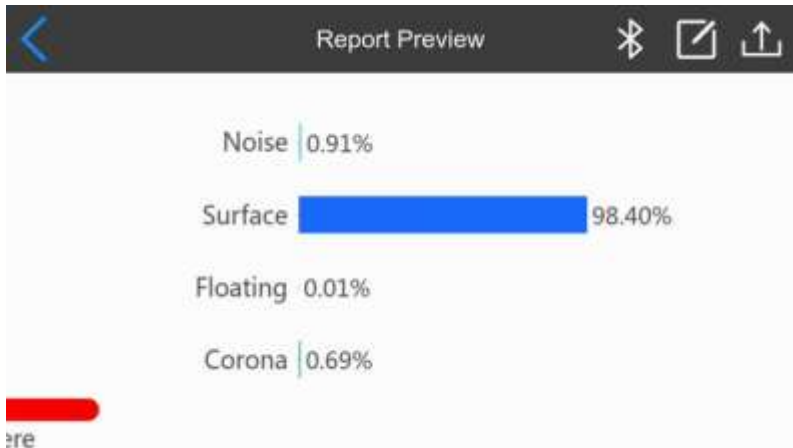
Report Generation



Voice Annotation



Bluetooth Share



Report Zoom

HERTZINNO

A Professional Company in Acoustic

CONTACT NOW

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Thank You



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info@hertzinno.com